

Assignment 5: Data Visualization

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Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

#1

```
library(tidyverse); library(here); library(cowplot)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.0      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
## here() starts at /home/guest/ENV 872/EDE_Fall2025
##
##
## Attaching package: 'cowplot'
##
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```
here()
```

```
## [1] "/home/guest/ENV 872/EDE_Fall2025"
```

```
PeterPaul.chem.nutrients <-
  read.csv(here("Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),
           stringsAsFactors = T)
Litter.Data <-
  read.csv(here("Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv"),
           stringsAsFactors = T)
#2
PeterPaul.chem.nutrients$sampldate <- ymd(PeterPaul.chem.nutrients$sampldate)
Litter.Data$collectDate <- ymd(Litter.Data$collectDate)
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
mytheme <- theme_classic(base_size = 12) +
  theme(axis.text = element_text(color = "red"),
        legend.position = "right",
        legend.text = element_text(size = 10), legend.title = element_text(size = 10))
theme_set(mytheme)
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (**tp_{ug}**) by phosphate (**po₄**), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the **lm** method. Adjust your axes to hide extreme values (hint: change the limits using **xlim()** and/or **ylim()**).

```
#4
theme_set(mytheme)

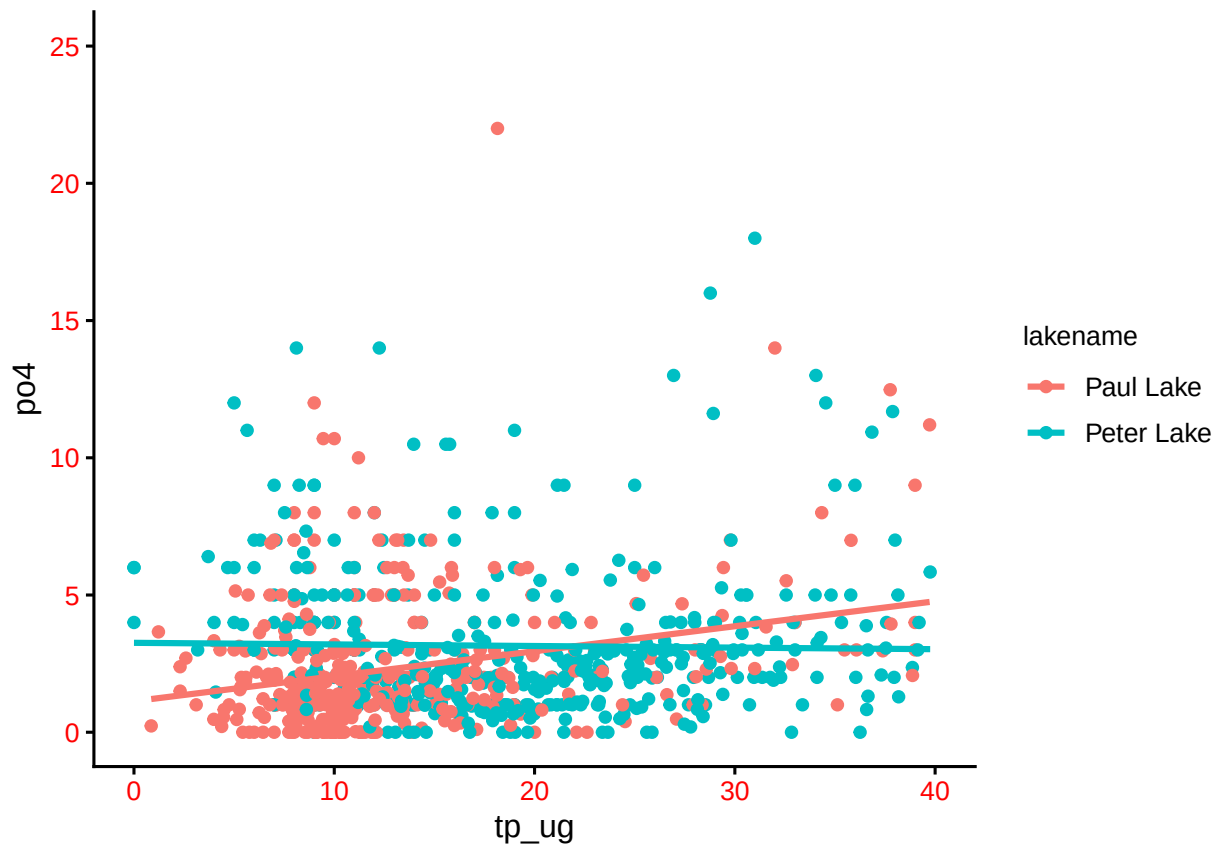
NTLtp_p <- PeterPaul.chem.nutrients %>%
  ggplot(aes(x = tp_ug,
             y = po4,
             color = lakename)) +
  geom_point() +
  geom_smooth(method = "lm",
             se = FALSE) +
  xlim(0, 40) +
  ylim(0, 25)

print(NTLtp_p)

## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 22122 rows containing non-finite outside the scale range
## ('stat_smooth()').

## Warning: Removed 22122 rows containing missing values or values outside the scale range
## ('geom_point()').
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

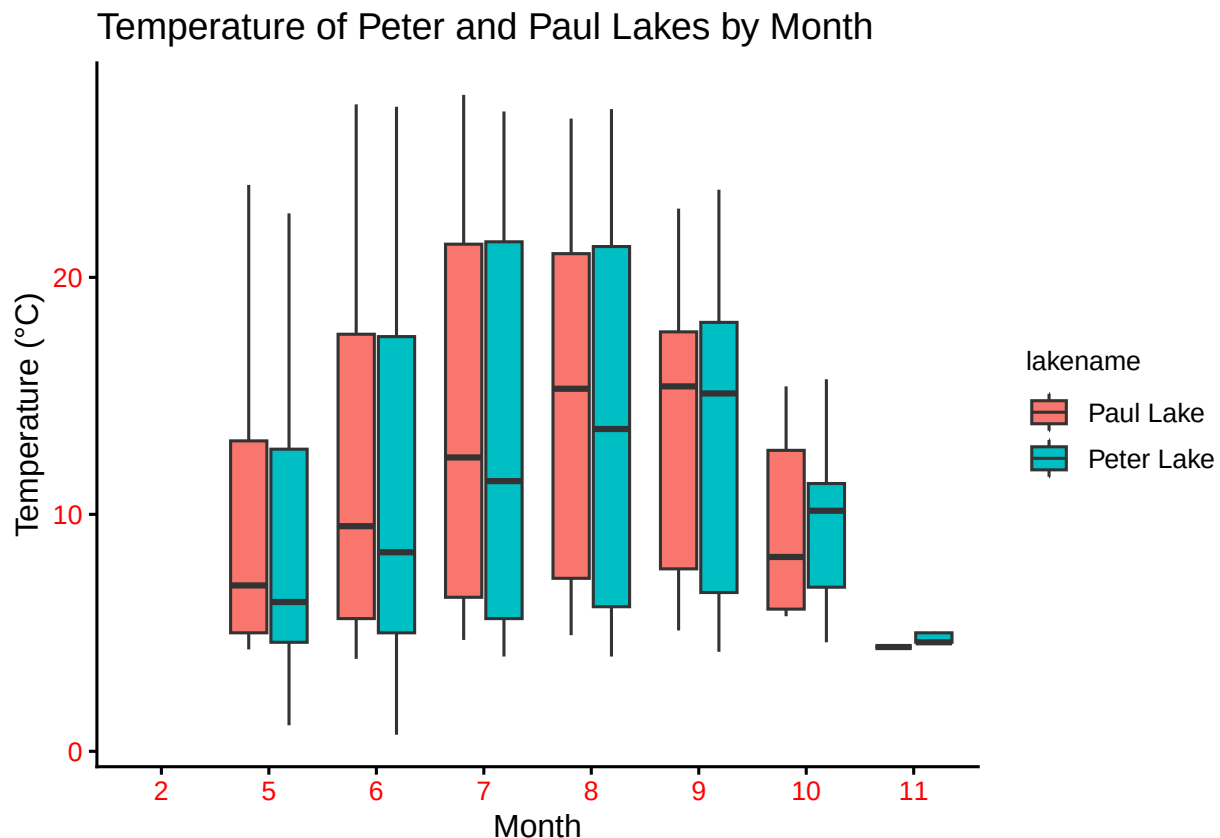
```
#5
theme_set(mytheme)

TempPeterPaul <- PeterPaul.chem.nutrients %>%
  ggplot(aes(x = factor(month),
             y = temperature_C,
             fill = lakename)) +
  geom_boxplot() +
  labs(title = "Temperature of Peter and Paul Lakes by Month",
       x = "Month",
       y = "Temperature (°C)",
       color = "Lake")

print(TempPeterPaul)
```

```
## Ignoring unknown labels:
## * colour : "Lake"
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



```

TpPeterPaul <- PeterPaul.chem.nutrients %>%
  ggplot(aes(x = factor(month),
             y = tp_ug,
             fill = lakename)) +
  geom_boxplot() +
  labs(title = "Phosphorus of Peter and Paul Lakes by Month",
       x = "Month",
       y = "Phosphorus",
       color = "Lake") +
  ylim(0, 75)

print(TpPeterPaul)

```

```

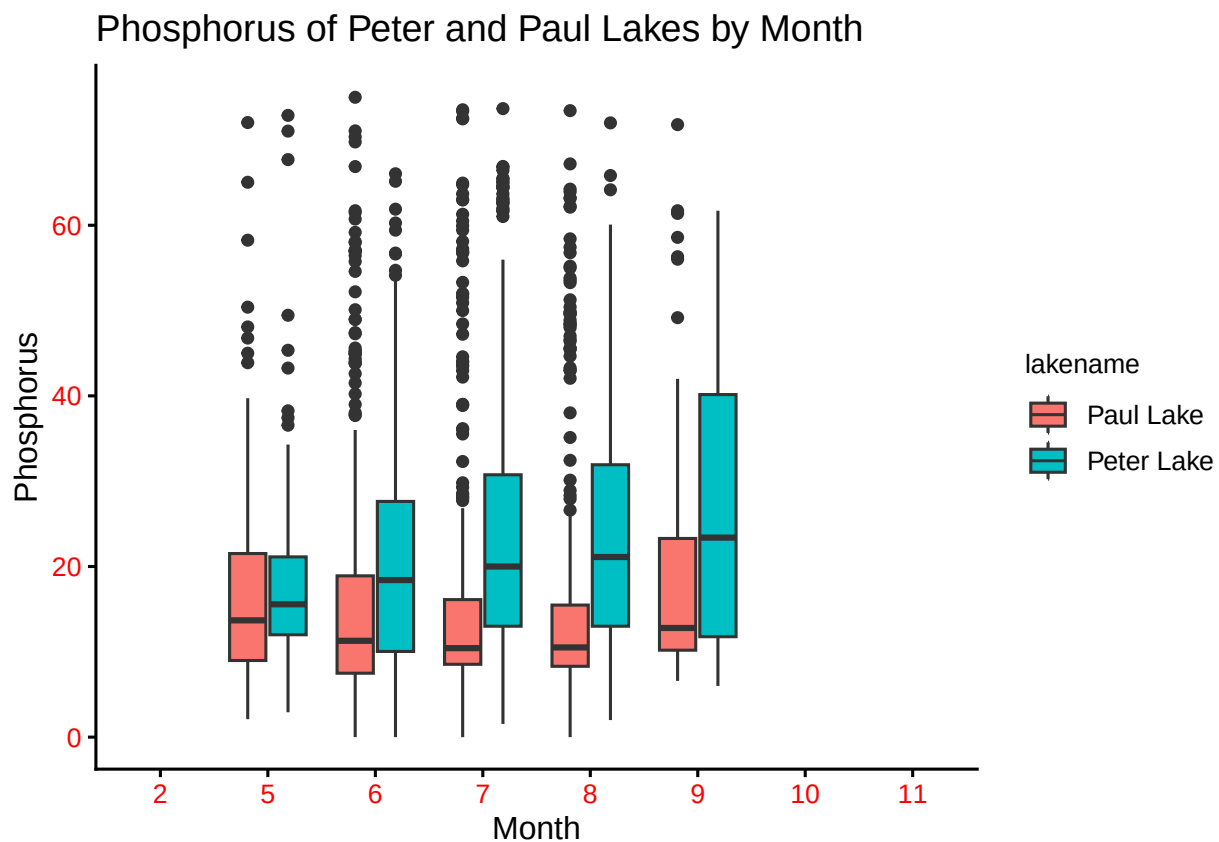
## Ignoring unknown labels:
## * colour : "Lake"

```

```

## Warning: Removed 20841 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```



```

TnPeterPaul <- PeterPaul.chem.nutrients %>%
  ggplot(aes(x = factor(month),
             y = tn_ug,
             fill = lakename)) +
  geom_boxplot() +

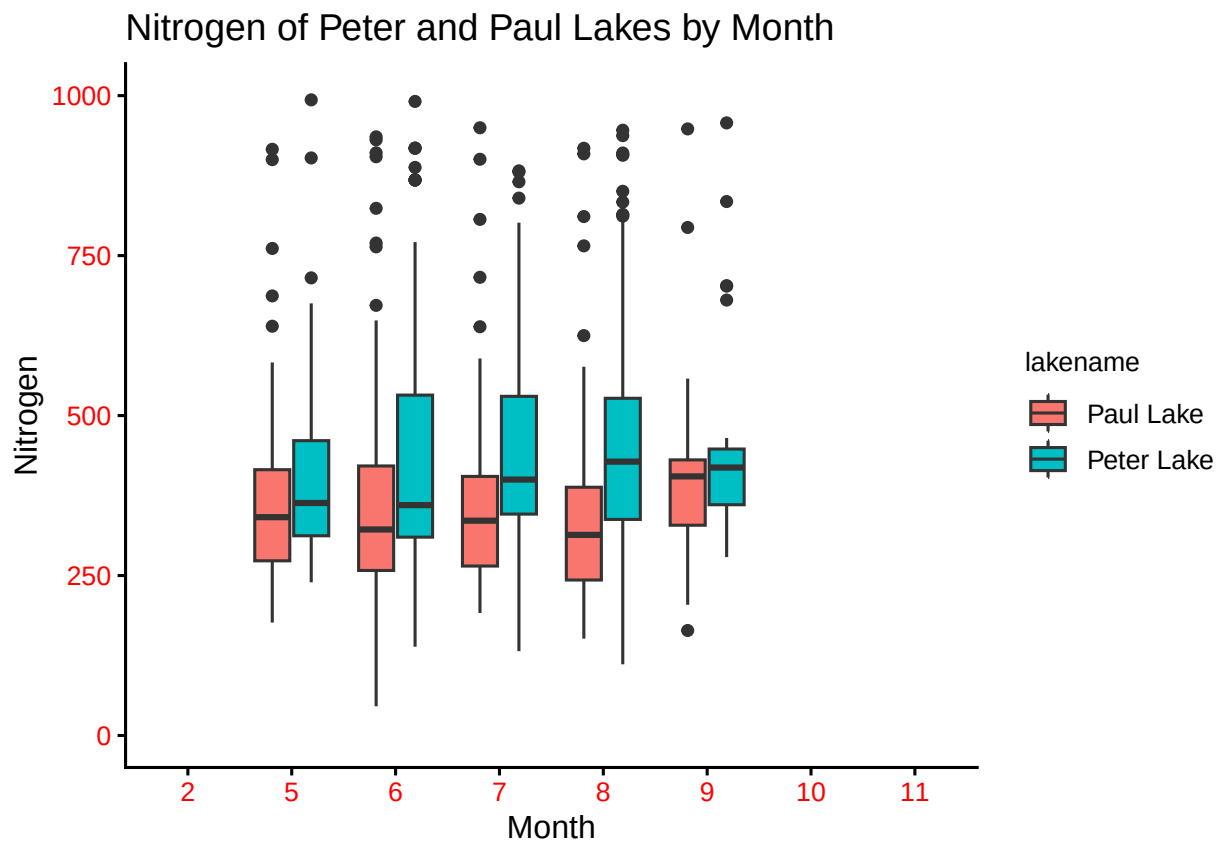
```

```
labs(title = "Nitrogen of Peter and Paul Lakes by Month",
     x = "Month",
     y = "Nitrogen",
     color = "Lake") +
ylim(0, 1000)

print(TnPeterPaul)
```

```
## Ignoring unknown labels:
## * colour : "Lake"
```

```
## Warning: Removed 21730 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



```
# Combining all three
no_x_axis_or_legend <- theme(
  axis.title.x = element_blank(),
  legend.position = "none")

TempPeterPaul <- TempPeterPaul + no_x_axis_or_legend
TpPeterPaul <- TpPeterPaul + no_x_axis_or_legend

move_x_axis <- theme(
  legend.position = "bottom")
```

```
TnPeterPaul <- TnPeterPaul + move_x_axis
```

```
TempTpTnPeterPaul <- plot_grid(TempPeterPaul, TpPeterPaul, TnPeterPaul, nrow = 3, align = 'h', rel_heights = c(1, 1, 1))
```

```
## Ignoring unknown labels:
## * colour : "Lake"
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

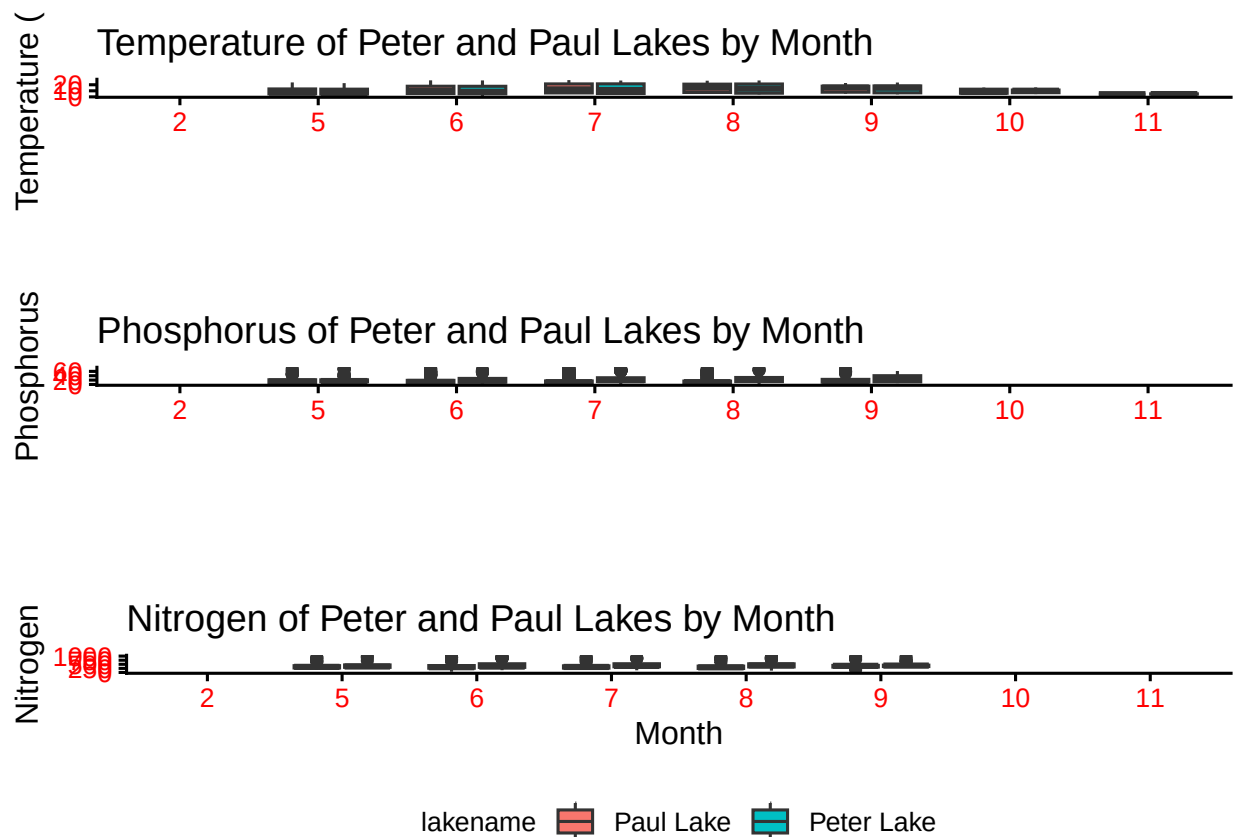
```
## Ignoring unknown labels:
## * colour : "Lake"
```

```
## Warning: Removed 20841 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Ignoring unknown labels:
## * colour : "Lake"
```

```
## Warning: Removed 21730 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
print(TempTpTnPeterPaul)
```



Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: Temperature has an increased value over the summer months, decreasing as you get closer to the end of the year. The average temperature between each lake is very similar, with Peter Lake being slightly colder in most months. Phosphorus and nitrogen are relatively consistent over the seasons, with Paul Lake having lower average concentrations of both in every observed month.

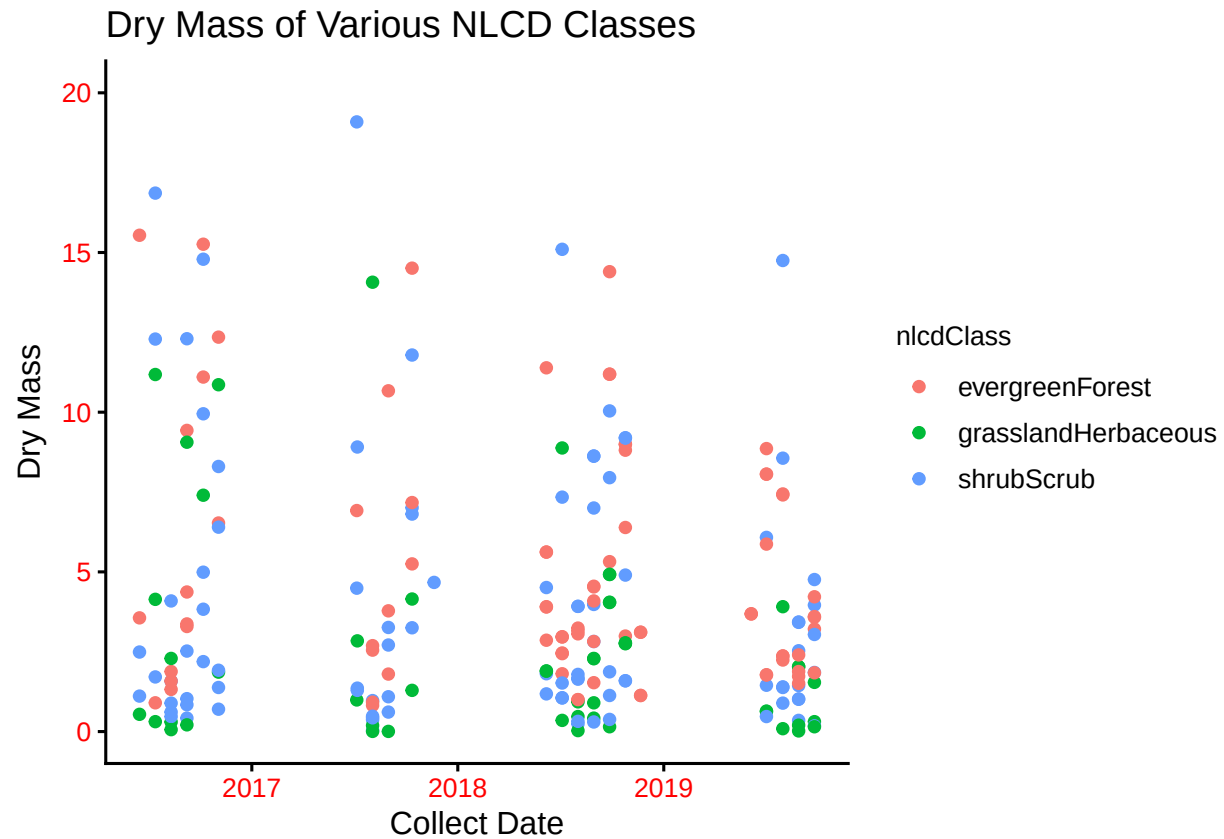
6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6
theme_set(mytheme)

NeedleLitterPlot <- ggplot(subset(Litter.Data, functionalGroup == "Needles"),
  aes(x = collectDate,
      y = dryMass,
      color = nlcdClass)) +
  geom_point() +
  ylim(0, 20) +
  ylab(expression("Dry Mass")) +
  xlab(expression("Collect Date")) +
  labs(title = "Dry Mass of Various NLCD Classes")

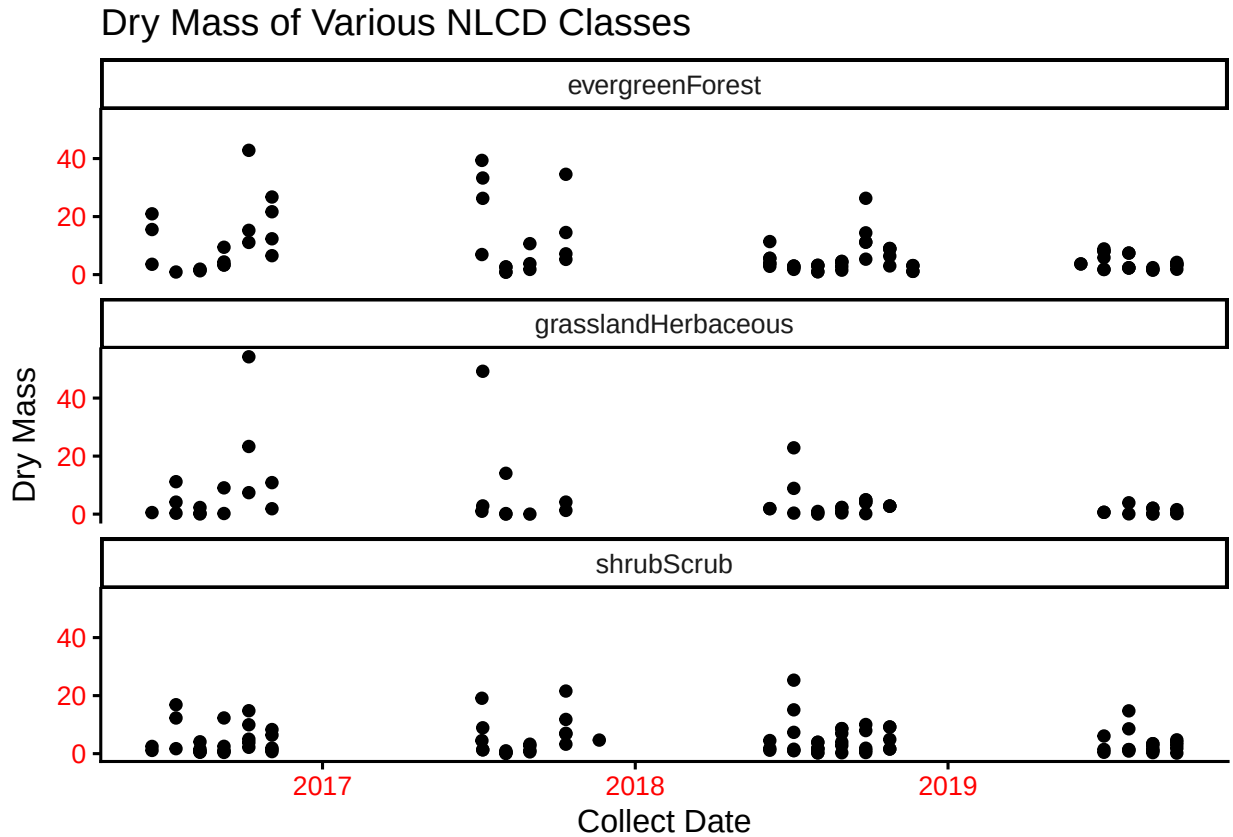
print(NeedleLitterPlot)
```

Warning: Removed 15 rows containing missing values or values outside the scale range
('geom_point()').



```
#7
FacetNeedleLitterPlot <- ggplot(subset(Litter.Data, functionalGroup == "Needles"),
  aes(x = collectDate,
      y = dryMass)) +
  geom_point() +
  facet_wrap(vars(nlcdClass), nrow = 3) +
  ylab(expression("Dry Mass")) +
  xlab(expression("Collect Date")) +
  labs(title = "Dry Mass of Various NLCD Classes")

print(FacetNeedleLitterPlot)
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: I think that plot 7 is more effective, because having separate plots makes it easier to distinguish the values of each NLCD class from each other. The colors are helpful, but it is easier to identify a trend when each graph is dedicated to one NLCD class. Plot 6 is more cluttered as well, which also makes it harder to understand what the average value might be for each class.